Project Documentation

Global Aviation Industry Analysis 2010–2026

30 Airlines × 17 Years Financials | Boeing vs Airbus vs COMAC | Regional Passenger Traffic | Global Routes | Aviation Incidents

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13.1 Project Overview

The **Global Aviation Industry Analysis 2010–2026** project analyzes major trends in the global aviation industry using financial, passenger, aircraft-order, route, and aviation-incident data.

The project covers:

- 30 major airlines
- 2010–2026 industry trends
- Airline financial performance
- Passenger demand
- Aircraft orders and deliveries
- Boeing vs Airbus competition
- COMAC C919 growth
- Global route performance
- COVID-19 impact and recovery
- Major aviation incidents
- Middle East aviation expansion

The project was developed as an end-to-end analytics workflow using **PostgreSQL, SQL, Excel, Power Query, Power BI, and DAX**.

13.2 Business Objective

The primary objective was to understand how the global aviation industry changed between **2010 and 2026** and identify important business trends, risks, and opportunities.

The analysis focused on five major areas:

1. **Airline financial performance**
2. **Aircraft manufacturer competition**
3. **Passenger traffic and regional recovery**
4. **Route economics**
5. **Aviation incidents**

13.3 Dataset Overview

The project contains five CSV datasets.

Dataset	Rows	Purpose
airline_financials.csv	497	Airline financial and operational performance
fleet_orders.csv	86	Aircraft orders, deliveries and backlog
passenger_traffic.csv	1,176	Regional monthly passenger traffic
route_performance.csv	400	Major global route performance
aviation_incidents.csv	40	Selected major aviation incidents

13.4 Technology Stack

The project used the following tools:

CSV Files

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PostgreSQL

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SQL

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Excel

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Power Query

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Power BI

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DAX

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Interactive Dashboard

Tools

- **PostgreSQL** — database storage, cleaning and SQL analysis
- **Excel** — exploratory analysis and validation

- **Power Query —** data transformation and preparation
- **Power BI —** data modeling and visualization
- **DAX —** analytical measures and time intelligence

Power BI's modeling guidance recommends separating dimensions used for filtering/grouping from fact tables used for summarization; one-to-many relationships are a common pattern for this structure.

13.5 PostgreSQL Implementation

The CSV datasets were imported into PostgreSQL and converted into structured database tables.

Main tables

airline_financials

fleet_orders

passenger_traffic

route_performance

aviation_incidents

PostgreSQL tasks performed

- Created tables
- Defined appropriate data types
- Imported CSV data
- Created primary keys where appropriate
- Validated row counts
- Checked NULL values
- Checked duplicate records
- Checked invalid numeric values
- Cleaned inconsistent text
- Performed SQL analysis

13.6 SQL Analysis

SQL was used to answer analytical questions such as:

- Revenue by airline
- Revenue by year

- Passenger trends
- Operating-margin analysis
- Manufacturer orders
- Aircraft deliveries
- Regional traffic
- Route rankings
- Incident analysis

Advanced SQL concepts used included:

GROUP BY

HAVING

CASE

JOIN

Subqueries

CTEs

Window Functions

RANK

DENSE_RANK

ROW_NUMBER

COALESCE

Date Functions

Aggregations

13.7 Excel Analysis

Excel was used as an intermediate analytical and validation tool.

Activities performed

- Data validation
- PivotTables
- Pivot Charts
- Filtering
- Sorting
- KPI analysis

- Trend analysis
- Cross-checking results

Excel helped validate the data before building the final Power BI model.

13.8 Power Query

Power Query was used to prepare the data before loading it into the Power BI model.

Transformations included

- Data type conversion
- Column renaming
- Removing unnecessary columns
- Handling missing values
- Removing duplicates
- Text cleaning
- Date transformation
- Creating the Date dimension
- Preparing analytical tables

Power BI uses Power Query to transform and prepare source data before it becomes part of the semantic model.

13.9 Power BI Data Model

The Power BI model uses a **single Dim Date table** as the shared time dimension.

The main analytical tables are connected to the date dimension through date/year fields according to the project's model design.

Simplified model

```
┌─────────────┐
│ Dim Date │
└─────┬─────┘
   │
┌──────────┼──────────┐
│   │   │
▼   ▼   ▼
```

```
┌──────────────┐ ┌──────────────┐ ┌───────────────┐
│Airline      │ │Fleet Orders  │ │Passenger Traffic│
│Financials    │ │         │ │        │
└──────────────┘ └──────────────┘ └───────────────┘
   │
   │
   ├───────────────────┐
   ▼         ▼
┌────────────────┐ ┌─────────────────┐
│Route Performance │ │Aviation Incidents │
└────────────────┘ └─────────────────┘
```

The model uses **one-to-many relationships** from the date dimension toward the analytical tables where the date keys match the required grain.

A Power BI relationship propagates filters between related tables, allowing slicers such as Year to affect related analytical tables.

13.10 DAX Measures

The project contains reusable DAX measures.

Financial Measures

Total Revenue

Revenue Growth %

Revenue YTD

Airline Measures

Total Passengers

Average Load Factor

Airline Rank

Fleet Measures

Total Orders

Total Deliveries

Total Backlog

Passenger Measures

Total RPK

Total ASK

Route Measures

Route Revenue

Average Fare

Safety Measures

Total Incidents

Total Fatalities

Fatal Incidents

Geopolitical Incidents

Boeing Incidents

Airbus Incidents

Explicit DAX measures are useful because they allow calculations to respond dynamically to the filter context of Power BI visuals.

13.11 Power BI Dashboard

The final report contains **five analytical pages**.

Page 1 — Executive Overview

KPIs

- Total Revenue
- Total Passengers
- Average Load Factor
- Total Orders
- Total Deliveries
- Total Incidents

Visuals

- Revenue Trend
- Passenger Trend
- Revenue by Region
- Top 10 Airlines
- Year slicer

- Region slicer
- Airline slicer
- Business Model slicer

13.12 Page 2 — Airline Performance

This page analyzes airline-level performance.

KPIs

- Total Revenue
- Total Passengers
- Average Load Factor
- Revenue Growth %

Visuals

- Revenue by Airline
- Operating Margin by Airline
- Passengers by Airline
- Fleet Size by Airline
- Revenue by Business Model
- Revenue vs Passengers

Purpose

To identify high-performing airlines and compare different airline business models.

13.13 Page 3 — Boeing vs Airbus

This page focuses on aircraft manufacturers.

KPIs

- Net Orders
- Deliveries
- Backlog
- Manufacturer Count

Visuals

- Aircraft Orders by Manufacturer
- Aircraft Deliveries by Manufacturer

- Backlog by Manufacturer
- Orders by Aircraft Family
- Boeing vs Airbus Order Trend
- Orders by Aircraft Category

Purpose

To analyze the competitive position of:

- Boeing
- Airbus
- COMAC
- Embraer

13.14 Page 4 — Passenger & Route Analysis

This page analyzes passenger demand and route performance.

KPIs

- Total RPK
- Total ASK
- Average Load Factor
- Route Revenue
- Average Fare

Visuals

- Regional Passenger Traffic Trend
- Regional Load Factor Trend
- Passenger Traffic by Region
- Top 10 Global Routes
- Top 10 Routes by Revenue
- Fare vs Passenger Volume

Purpose

To understand:

- COVID recovery
- Regional demand
- Route performance

- Passenger volume
- Fare economics

13.15 Page 5 — Aviation Safety

This page analyzes the selected major aviation incidents.

KPIs

- Total Incidents
- Total Fatalities
- Fatal Incidents
- Geopolitical Incidents

Visuals

- Incidents by Year
- Fatalities by Year
- Incidents by Severity
- Boeing vs Airbus Incidents
- Incidents by Airline
- Fatalities by Aircraft Type
- Geopolitical vs Operational Incidents

13.16 Key Business Insights

The analysis identified several important industry trends.

COVID-19

COVID-19 caused a dramatic decline in passenger demand.

Asia Pacific experienced the strongest modeled decline, with RPK falling approximately **86%** between December 2019 and April 2020.

Boeing 737 MAX

Boeing's 737-family net orders fell dramatically during the MAX crisis and COVID period.

2018 → +675

2019 → -87

2020 → -1,026

2024 → +569

Airbus

Airbus showed stronger narrowbody order momentum during the post-MAX period.

COMAC

COMAC's C919 has moved from development toward commercial production, particularly in China's domestic market.

Middle East

The Middle East has become an increasingly important global aviation hub, driven by airlines such as Emirates, Qatar Airways, Etihad, Saudia and Riyadh Air.

Route Economics

Passenger volume alone is insufficient for route evaluation. Fare, revenue, distance, frequency and operating cost should also be considered.

13.17 Business Recommendations

Recommendation 1

Monitor Boeing and Airbus using:

Orders

Deliveries

Backlog

Narrowbody share

Widebody share

Recommendation 2

Prioritize capacity planning according to regional passenger recovery.

Recommendation 3

Evaluate routes using revenue **and** cost rather than passenger volume alone.

Recommendation 4

Monitor operating margins alongside revenue growth.

Recommendation 5

Track COMAC as an emerging competitor in the narrowbody market.

Recommendation 6

Monitor Middle Eastern airlines as major competitors for global connecting traffic.

13.18 Project Limitations

The project has several limitations.

Modeled Data

Some values were interpolated or modeled rather than directly sourced for every year.

Financial Data

Foreign airline revenue was converted to USD and may be affected by exchange rates.

Route Data

Route passenger and revenue values are estimates rather than complete schedule-level commercial datasets.

Incident Data

Only 40 selected major incidents are included.

2026

The 2026 dataset represents a **partial year** according to the project methodology.

Safety Analysis

The incident dataset should not be interpreted as a complete measure of global aviation safety.

Profitability

Route revenue is available, but complete fuel, crew, airport, maintenance and aircraft operating costs are not included. Therefore, the route analysis is primarily a **revenue attractiveness analysis**, not a full profitability model.

13.19 Final Project Conclusion

The Global Aviation Industry Analysis 2010–2026 project demonstrates an end-to-end data analytics workflow using PostgreSQL, SQL, Excel, Power Query, Power BI and DAX. The analysis examined airline financial performance, passenger demand, aircraft manufacturer competition, route economics and major aviation incidents.

The results highlight the significant impact of COVID-19, the competitive consequences of the Boeing 737 MAX crisis, Airbus's strong narrowbody position, the emergence of COMAC, and the growing strategic importance of Middle Eastern aviation hubs.

The project demonstrates how multiple datasets can be integrated into a single analytical model and transformed into interactive dashboards and actionable business insights.

13.20 Skills Demonstrated

This project demonstrates practical experience with:

SQL

PostgreSQL

Data Cleaning

Data Validation

Excel

PivotTables

Power Query

Power BI

Data Modeling

Star Schema

Relationships

DAX

Time Intelligence

Data Visualization

KPI Development

Business Analysis

Analytical Storytelling

📌 Final Project Structure

Global Aviation Industry Analysis

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├── 03_PostgreSQL_SQL

├── 04_Excel_Analysis

├── 05_Power_Query

├── 06_PowerBI_Data_Model

├── 07_DAX_Measures

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├── 08_Dashboard
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```